

Smart Label Compliance Inspector

Problem Statement ID – **26108**

Problem Statement Title – **AI-Powered**

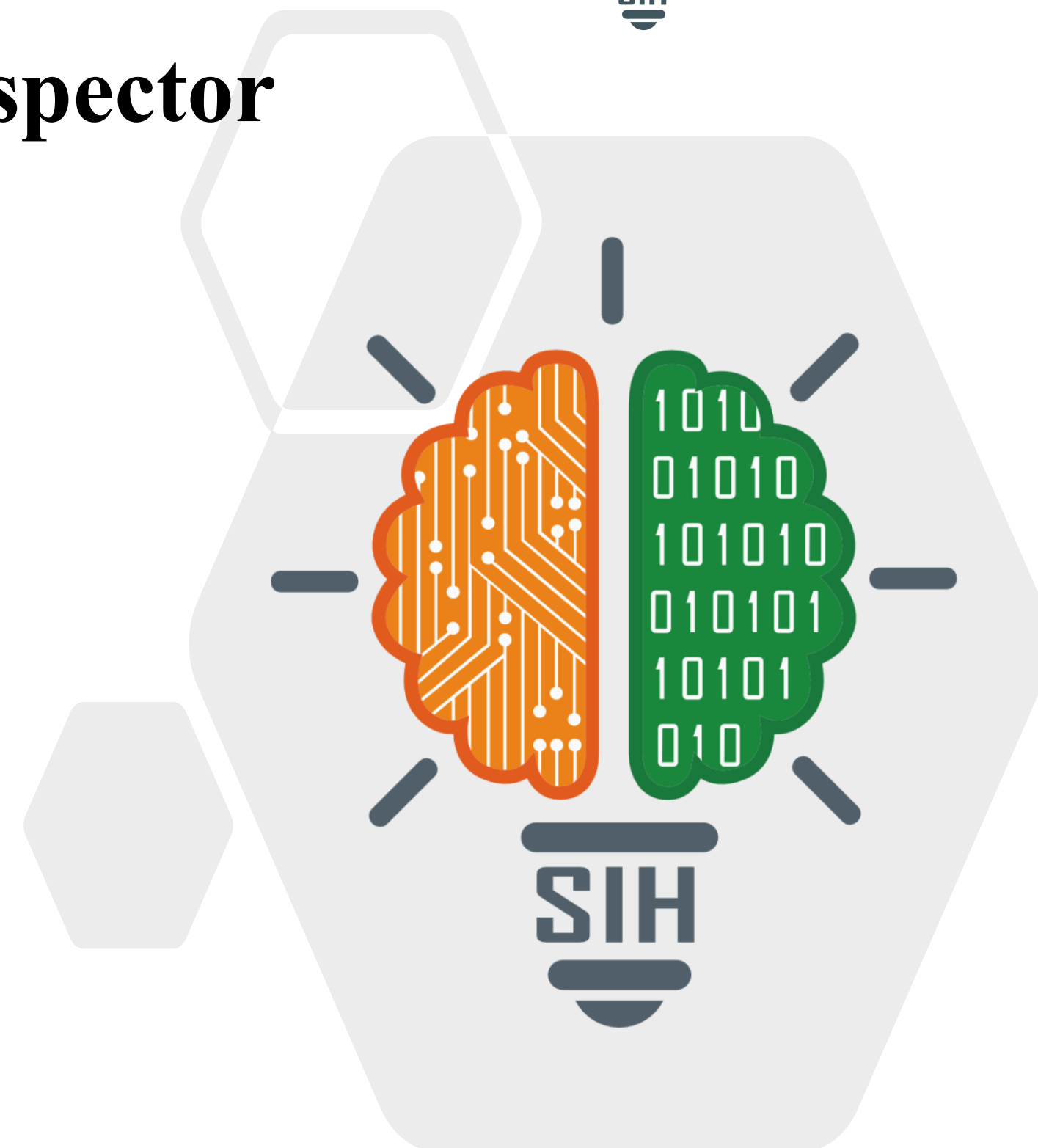
**Recommendation Engine for Identifying Applicable
Indian Standards for Procurement Specifications**

Theme – **Smart Automation**

PS Category – **Software**

Team ID – **149350**

Team Name: **N0TH1NG**





SMART LABEL COMPLIANCE INSPECTOR

AI-POWERED LEGAL METROSOLOGY VERIFICATION



SMART INDIA
HACKATHON
2026

PROPOSED SOLUTION



Multi-Input Specificity Intake:

Product description Technical specification Tender PDF/DOCX Portal API call



HybridSemantic Matching

Separates legal rules, vision extraction & evaluation



Calibration & Applicability

Real-world measurement via Artico/ID-1 reference



Date-Awares Compliance

Verdicts by packing date: Complaint / Non-Compliant / Indeterminate

HOW IT ADDRESSES THE PROBLEM



Objective Inspection

Replaces manual judgment with measured, evidence-based, reproducible results



Comprehensive Coverage

Automates Rule 6 (Mandatory text like MRP/Date) and physical checks under Rules 7 & 8 (Placement & Font Size)



Enforcement-Ready Reports

Delivers cited evidence and detailed findings for regulatory action

INNOVATION & UNIQUENESS



Physical-scale accuracy

Converts pixels into real measurements for font & size checks



Versioned Legal Rule Packs

Date-addressable, content hashed regulations for reproducible checks



Evidence-Grounded Extraction

Reports only values backed by a verified source region on the label



Uncertainty-Aware Verdicts

Flags INDeterminate when evidence is insufficient



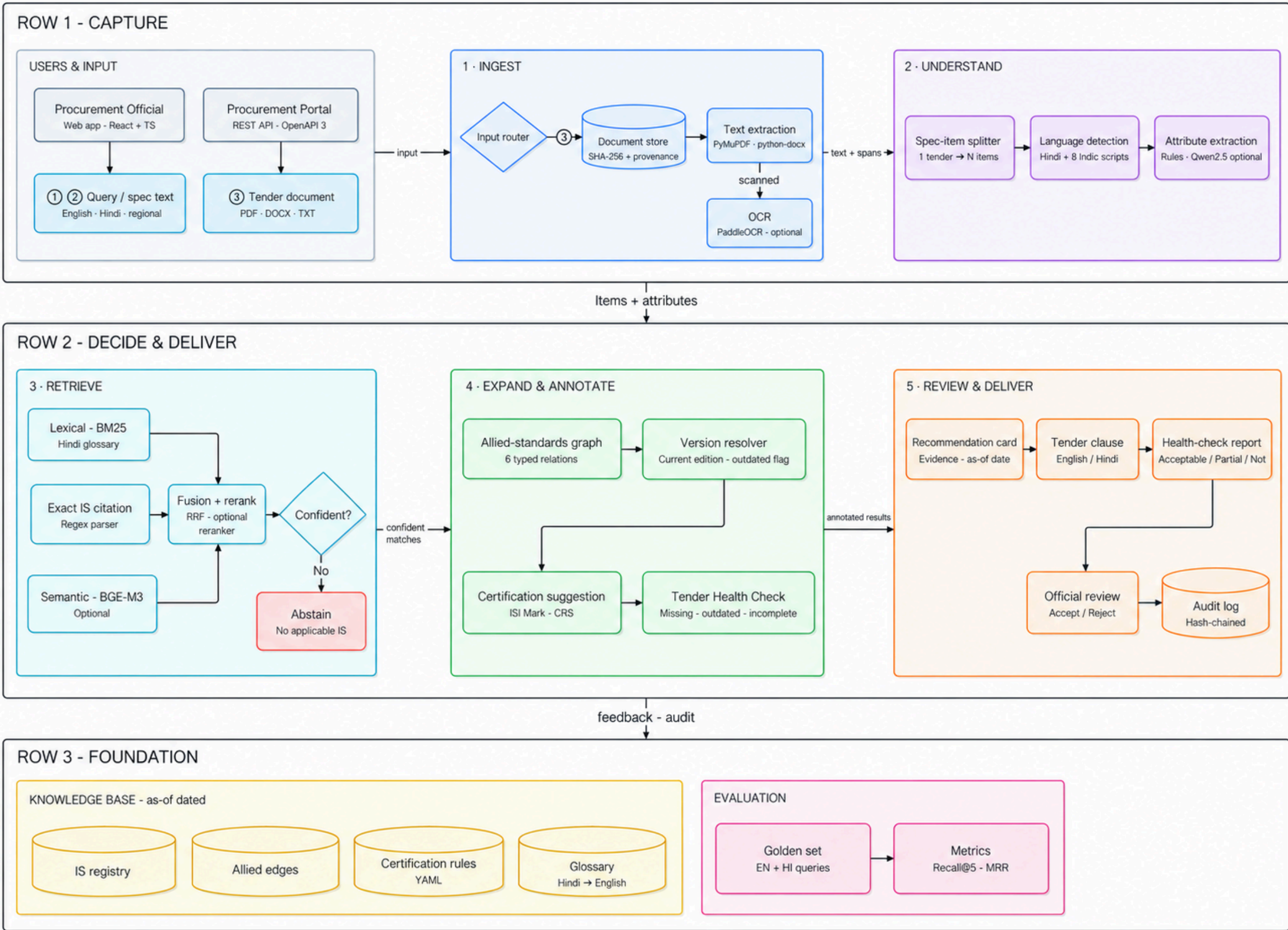
Clear Scope Handling

Distinguishes out-of-scope items from compliant products

TECHNICAL APPROACH

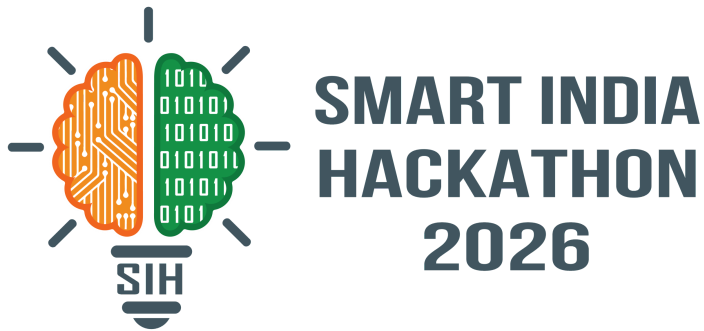
NOTH1NG

COMPONENT	TECHNOLOGIES
Document parsing	PyMuPDF + python-docx
OCR(scannedtenders)	PaddleOCR
Semantic search:	BGE-M3 + bge-reranker-v2-m3
Lexical search & fusion:	BM25 + Reciprocal Rank Fusion
LLM extraction	Qwen2.5 via Ollama or hosted API
Knowledge base	IS registry + allied graph (PostgreSQL + pgvector)
Rule packs & audit	Git + YAML . hash-chained audit log





FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

- 1. Working Prototype: 5-stage pipeline, web app, REST API and 12 automated tests are already built and running.
- 2. Mature Open-Source Stack: Runs on a CPU server; lexical mode works offline, and semantic mode is an optional upgrade.
- 3. Measurable Quality: Golden-set evaluation in CI (Recall@5, MRR, abstention accuracy); baseline Recall@5 of 1.00 on an internal 28-query set.
- 4. Phased Development: 5 seed domains → BIS registry sync → Semantic + rerank → Portal pilot (GeM/CPPP) → Analytics

Challenge	Mitigation
Access to standards data	Public BIS metadata; full texts only for demo set
Frequent revisions & amendments	Scheduled registry sync + version resolver
Overlapping scopes	Hybrid retrieval + reranker + ranked evidence
Services / no applicable IS	Confidence threshold with abstention
Hindi technical terms	BGE-M3 on original text + domain glossary

How the value flows

TARGET USERS



Procurement officials

Govt depts, PSEs, agencies,
private orgs



Portals (via API)

GeM · CPPP · State e-
procurement



Ecosystem

Department of Consumer
Affair · BIS · Manufacturers,
suppliers

Standards in seconds

IMPACT



Procurement Official — Standards for a spec in seconds
+ a defensible, ready-to-paste clause



Department / BIS — Fewer outdated citations + insight
into missed QCO/CRS requirements



Supplier — Clear, complete requirements → fewer
disputes and rejected bids



Citizen — Public assets bought to current safety &
quality standards

Social

Safer, better-quality
products

Economic

Fewer Disputes, Fairer
Procurement

Governance

Compliant, Auditable,
Human-Verified Decisions

Environmental

Less Waste, Fewer Re-
Procurements

Scalable

From 5 Domains to 22,700
Standards

1. Bureau of Indian Standards — Know Your Standard. [BIS — Know Your Standard](#)
2. Bureau of Indian Standards — Products under Compulsory Certification. [BIS — Compulsory Certification](#)
3. Bureau of Indian Standards — Compulsory Registration Scheme. [BIS — Registration Scheme](#)
4. Bureau of Indian Standards — Hallmarking of gold jewellery & artefacts. [BIS — Hallmarking](#)
5. BIS Standards Portal — Indian Standards texts. [BIS Standards Portal](#)
6. PIB — BIS notifies 187 Quality Control Orders. [PIB Press Release](#)
7. Department of Expenditure — General Financial Rules 2017. [Department of Expenditure](#)
8. Department of Expenditure — Manual for Procurement of Goods (June 2022). [Procurement Manual PDF](#)
9. Chen, J. et al. (2024) — BGE M3-Embedding. [arXiv:2402.03216](#)
10. Gala, J. et al. (2023) — IndicTrans2. [arXiv:2305.16307](#)
11. Cormack, G. V. et al. (2009) — Reciprocal Rank Fusion, SIGIR '09. [DOI: 10.1145/1571941.1572114](#)

Prototype data: public BIS metadata for 5 product domains, as of 30-09-2026; every record to be verified on the BIS portal

GITHUB LINK: <https://github.com/4bh7n4v/AI-Powered-Indian-Standards-Recommendation-Engine>